WORKED SOLUTION

STEP II 1993, Question 8

Pure • AM-GM Inequality by Induction

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s2-q8

This question has a full guided treatment in *Beyond STEP*, Section 3.2, *AM–GM, Backwards*: https://beyondstep.keithmcnulty.org/ch03/am-gm-backwards.html

The key idea

The two-term inequality is just a square being non-negative. To go from 2^m terms to 2^{m+1}, split the list into two halves of length 2^m, use the induction hypothesis on each half, and then use the two-term inequality on the two half-means. That proves AM–GM only when the number of terms is a power of 2; the final part fills the gap with a neat trick (due to Cauchy): pad the list with copies of its own mean, which changes neither the mean nor anything essential about the product.

Solution

Two terms

Expanding,

$$\left(\frac{a_1+a_2}{2}\right)^2 - a_1a_2 = \frac{a_1^2+2a_1a_2+a_2^2-4a_1a_2}{4} = \left(\frac{a_1-a_2}{2}\right)^2 \geq 0,$$

so $a_1a_2 \leq \left(\frac{a_1+a_2}{2}\right)^2$, with equality exactly when $a_1 = a_2$. Notice that this holds for *any* two positive numbers; we shall use it below for numbers other than a_1 and a_2.

The induction

Let $P(m)$ be the statement that $(*)$ holds for every list of 2^m positive numbers. The case $m = 1$ is the two-term inequality just proved.

Now suppose $P(m)$ holds, and take 2^{m+1} positive numbers $a_1, \ldots, a_{2^{m+1}}$. Write $N = 2^m$

and split them into two halves of N terms each, with means

$$X = \frac{a_1 + \cdots + a_N}{N}, \qquad Y = \frac{a_{N+1} + \cdots + a_{2N}}{N}.$$

Applying $P(m)$ to each half,

$$a_1 \cdots a_N \le X^N \qquad \text{and} \qquad a_{N+1} \cdots a_{2N} \le Y^N.$$

All the quantities here are positive, so we may multiply these two inequalities:

$$a_1 \cdots a_{2N} \le X^N Y^N = (XY)^N.$$

Now X and Y are positive, so the two-term inequality gives $XY \le \left(\frac{X+Y}{2}\right)^2$. Since $t \mapsto t^N$ is increasing for $t > 0$, raising both sides to the power N keeps the direction:

$$(XY)^N \le \left(\frac{X+Y}{2}\right)^{2N}.$$

Finally, $\dfrac{X+Y}{2} = \dfrac{(a_1 + \cdots + a_N) + (a_{N+1} + \cdots + a_{2N})}{2N}$ is the mean of all $2N = 2^{m+1}$ numbers. Putting the inequalities together,

$$a_1 \cdots a_{2^{m+1}} \le \left(\frac{a_1 + \cdots + a_{2^{m+1}}}{2^{m+1}}\right)^{2^{m+1}},$$

which is $P(m+1)$. By induction, $(*)$ holds for all positive integers m.

Ask yourself Is it legitimate to multiply two inequalities together like this?

If $0 < p \le q$ and $0 < r \le s$ then $pr \le qr \le qs$, using positivity of r and then of q. Without positivity this can fail (for example $-3 \le 1$ and $-3 \le 1$, but $9 > 1$), which is why the condition $a_i > 0$ matters.

Padding to a power of 2

The printed definition has $b_1 = a_2$, which must be a misprint for $b_1 = a_1$: we take $b_i = a_i$ for $1 \le i \le n$ and $b_i = A$ for $n < i \le 2^m$. Each b_i is positive (since A is a mean of positive numbers). Their sum is

$$b_1 + \cdots + b_{2^m} = (a_1 + \cdots + a_n) + (2^m - n)A = nA + (2^m - n)A = 2^m A,$$

so the mean of the b_i is exactly A. Their product is $a_1 \cdots a_n \, A^{2^m - n}$. Applying $(*)$ to the b_i therefore gives

$$a_1 \cdots a_n \, A^{2^m - n} \le A^{2^m},$$

as required.

The AM–GM inequality

Given any positive integer n, choose m with $2^m > n$ (possible since $2^m \to \infty$), so the previous step applies. Dividing by A^{2^m-n}, which is positive, gives

$$a_1 \cdots a_n \leq A^n.$$

Both sides are positive and $t \mapsto t^{1/n}$ is increasing on $t > 0$, so taking nth roots,

$$(a_1 \cdots a_n)^{1/n} \leq A = \frac{a_1 + \cdots + a_n}{n},$$

which is the AM–GM inequality. (If n happens to be a power of 2 we could instead quote $(*)$ directly, but choosing $2^m > n$ means no separate case is needed.)

Checks

Try $n = 3$ with $a = (1, 2, 6)$: $A = 3$, and with $m = 2$ the padded list is $(1, 2, 6, 3)$, whose mean is 3 and whose product is $36 \leq 3^4 = 81$; dividing by 3 gives $12 \leq 27 = A^3$, so $\sqrt[3]{12} \approx 2.29 \leq 3$. Also, when all the a_i are equal every inequality used becomes an equality, as it should. A computer testing $(*)$ for $m \leq 6$ and the final inequality for $n \leq 40$ on thousands of random lists, in exact rational arithmetic, found no failures.

Common slips

- Applying the induction hypothesis to all 2^{m+1} terms at once: it only covers lists of length 2^m, so you must split into halves.
- Forgetting to check that the padded list still has mean A; this is the whole point of padding with A rather than anything else.
- Taking nth roots or dividing without noting that the quantities are positive.

Beyond STEP

This "forward–backward" induction is Cauchy's proof of AM–GM: prove it for powers of 2, then come back down to every n. AM–GM is also a special case of Jensen's inequality for the concave function ln, and a whole family of power-mean, Hölder and Minkowski inequalities grows out of the same ideas; these are basic tools in first-year analysis. Bartle and Sherbert, *Introduction to Real Analysis*, uses inequalities like these from the start, and Siklos, *Advanced Problems in Mathematics*, has more STEP-style practice.

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